Soren Larsen

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Summary

Product and platform engineer with an M.S. in NLP who joins early-stage teams during zero-to-one development, working directly with customers through prototypes, implementation, launch, and production support. Shipped enterprise onboarding, billing, integrations, deployment, and cloud infrastructure while helping scale an AI-agent platform from fewer than five users to hundreds.

Education

University of California, Santa Cruz Sep 2024 – Dec 2025

Master of Science, Natural Language Processing *Santa Cruz, CA*

University of California, Santa Cruz Sep 2019 – Jun 2023

Bachelor of Science, Computer Science *Santa Cruz, CA*

Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, Bash, Dart
- **AI / NLP:** LLM agents, LangGraph, MCP, Ollama, RAG, tool calling, prompt engineering, retrieval and agent evaluation, LLM-as-judge, multi-agent systems, observability, information extraction, knowledge graphs, multimodal AI, Hugging Face Transformers, PyTorch, sentence-transformers, FAISS
- **Full-Stack / Infra:** FastAPI, REST APIs, webhooks, OAuth, authentication, PostgreSQL, Firebase, multi-tenancy, billing systems, React, Next.js, Flutter, Docker, GCP, GitHub Actions, CI/CD, release tooling with rollback, automated testing, production debugging
- **Solutions Engineering:** Requirements gathering, technical scoping, rapid prototyping, API integrations, implementation, production support

Experience

Contract engineering

Founding Engineer (Contract) Feb 2026 – Sep 2026

Levangie Laboratories *Two-engineer AI-agent platform team, Remote*

Joined after a Gray Whale–Levangie pilot. The contract ended once the platform build was finished.

- **Customer Acquisition:** Worked directly with a customer from requirements and prototype demos through launch, turning an open-source web chat widget into an embeddable production AI assistant with streaming responses, tenant-isolated memory, and abuse controls that unblocked the pilot and purchase and was later deployed for additional customers.
- **Enterprise Onboarding:** Built the multi-tenant workspace and billing system that enabled the first organizational customer, including centralized access, shared billing and memory, and payment, refund, and transfer workflows.
- **Cloud Runtime:** Built and load-tested a replacement GCP/Docker runtime for on-demand isolated agent containers, eliminating capacity waits and crashes as the platform grew from fewer than five users to hundreds.
- **Integrations Platform:** Built multi-account OAuth, webhooks, scheduling, and in-chat setup for session-aware agent workflows, replacing a customer's file-based workaround and adding persistent memory and automated reconciliation.
- **Release Engineering:** Built four GitHub Actions workflows and a one-command release tool with rollback, replacing manual deployments with tested promotion across development, staging, and production.
- **Incident Response:** Resolved a production cascade that left users unable to load agents by fixing blocking authentication work and deferring heavy startup imports; cut API startup memory by 75%, from 594 MB to 147 MB, and helped add circuit breakers, rate limits, and monitoring that prevented recurrence.

AI Systems Consultant (Contract) Jun 2025 – Jan 2026

Gray Whale *Four-engineer AI team, Remote*

- **Multi-Agent Prototype:** Built a LangGraph planner-executor around a local Hugging Face/PyTorch model, with dynamic agent discovery and a wrapper adopted by all three co-engineers for 5–10 specialized agents.
- **Developer Enablement:** Built a tool that converted participants' datasets into the required JSON format, enabling hands-on projects at Developer Camp, an AI hackathon.
- **AI Evaluation:** Built BM25, sentence-transformers, and FAISS baselines for a 700-document benchmark; found the semantic systems could not be meaningfully separated without a much larger corpus, informing a shift to controlled tool-routing evaluation.
- **Real-Time Transcription:** Built a native audio-filter plugin for a desktop streaming application that ran speech to text on-device with whisper.cpp, using CoreML and Metal acceleration on Apple Silicon.
- **Adaptive Automation:** Prototyped a scraper that used an LLM to regenerate CSS selectors when page markup changed, keeping extraction working across site redesigns.

Before and during graduate school (M.S., UC Santa Cruz)

ML/AI Engineer (Part-Time) May 2025 – Sep 2025

Onda Sports *Surf-forecasting app; sole engineer for AI assistant, Remote*

- **Production AI:** Took sole ownership of the assistant inside an externally developed iOS/Android codebase, owning its prompts, retrieval behavior, evaluation, and production integration while working directly with founders.
- **Observability:** Instrumented the assistant so the team could trace prompts, retrieved forecasts, model responses, latency, and token cost across releases.
- **Product Quality:** Built a roughly 70-prompt regression set that exposed incorrect skill-level interpretation despite accurate forecasts; redesigned and shipped retrieval and prompting behavior, verifying response and document-retrieval accuracy.

Full-Stack Developer and Product Consultant (Contract) Apr 2024 – Jun 2025

Boardal *Surfboard marketplace; first hire and only non-founder engineer, San Diego, CA*

- **User Engagement:** Increased daily active usage and brought users back to a marketplace with 20,000+ registered users by building saved-search alerts and push-notification delivery.
- **Mobile Ownership:** Shipped production messaging and listing workflows by owning delivery from scoping and Flutter UI through Firebase data flows, notifications, and iOS releases alongside a cofounder.

Selected Projects

No-RAGrets — M.S. NLP Capstone with CarbonBridge 2025

Industry-sponsored scientific literature review system

- **Discovery and Partnership:** Secured CarbonBridge as the industry sponsor and converted researchers' manual paper-review workflow into requirements for literature triage, extraction, and source verification through direct observation.
- **Technical Delivery:** Led architecture on a three-person team and built the multimodal extraction pipeline, Dgraph knowledge graph, and React interface for 48 papers and roughly 1,000 pages, allowing researchers to verify every extracted claim at its source.
- **Data Integrity:** Repaired 1,073 orphaned graph relationships without rebuilding the corpus by tracing a document-identity defect across the text and vision pipelines and writing a targeted migration.

local-agent-bench — Local Tool-Using Agent Benchmark 2026

Personal open-source project

- **Agent Evaluation:** Built a fully local agent with web search, page reading, and calculator tools on LangGraph, MCP, and Ollama, then benchmarked five open models on 15 tasks with three runs each; gpt-oss 20B passed 43 of 45 runs.
- **Failure Analysis:** Found that most models stalled after searching without opening the page to read the answer, measured prompt-injection resistance, and showed that hint sentences in tool descriptions helped some models and hurt others.

SLAC Accelerator Fault Investigator — Flower Hackathon at Stanford 2026

Team hackathon project on real SLAC accelerator data

- **Specification:** Wrote the design spec and 14 GitHub issues that the team's coding agents built from during the event.
- **Federated Agents:** Designed three instrument agents that each keep their own slice of data and send only summaries, under 1% of raw data, to an orchestrator that returns a verdict for human review; hosted the live demo after the event.

Sentinel — Cyber Incident Response Council, UCSC AI Hackathon 2026

Team project in the Local AI track; owned the Compliance agent

- **Compliance Agent:** Built the agent that works out which breach notification rules an incident triggers (GDPR, SEC, HIPAA, NIS2 and others) and how long is left on each deadline, with deadlines and human-escalation rules computed in code rather than by the model.
- **Model Selection:** Swept local models on the same cases and picked gemma4 26B, which passed 21 of 21 ATT&CK-based cases and 25 of 25 cases modeled on public breaches, all on one ASUS GX10 with no cloud calls.

team-build — Team Coordination Plugin for Coding Agents 2026

Personal open-source project

- **Agent Coordination:** Built a Claude Code and Codex plugin that keeps a team in sync when each person drives their own coding agent on one repo, storing shared decisions and API contracts in repo files every agent reads, with hooks and a sync check that run in both tools.

policy-tree — Medical Policy to Rule Tree Pipeline 2026

Personal open-source project

- **Document Pipeline:** Built a crawler that downloads an insurer's public medical-policy PDFs with retries and throttling, cuts out the initial criteria, and has an LLM turn them into nested AND/OR rule trees validated against a Pydantic schema.
- **Failure Handling:** Stored failed policies with their errors so they show up in the web UI instead of disappearing; runs on OpenAI or a local Ollama model, with 33 offline tests.

repo-impact — Engineering Impact Dashboard for Any GitHub Repo 2026

Personal open-source project, live demo

- **Impact Scoring:** Built a one-page dashboard that scores substantive reviews, changes to files other engineers also work in, and human-reviewed work instead of PR, commit, or line counts, with every number linking to the formula and inputs behind it.
- **Data Completeness:** Fetched every merged PR in the window one day at a time to get past GitHub's 1,000-result search cap, checked the counts match, and excluded bots; on PostHog over two weeks, 41% of merged PRs had only a bot approval.

StockPulse — Stock Trend and Sentiment App 2026

Personal project, deployed publicly on AWS

- **Public LLM App:** Built a stock trend app with FastAPI and React, using gpt-oss 20B on Groq to score sentiment from Reddit, ApeWisdom, and news feeds.
- **AWS with Terraform:** Moved it from Render to AWS Lambda as a container image, with DynamoDB for the cache and history, the API key in SSM, and every resource in Terraform; GitHub Actions deploys through OIDC, so no AWS keys are stored.
- **Cost Controls:** The app meters its own Lambda usage and switches itself off at 80% of the free tier; per-IP rate limits and a daily model-call cap shared across instances protect the Groq quota.

Ranked AI League — Browser Soccer Game 2026

Personal project

- **Full-Stack Delivery:** Built a Supabase-backed browser soccer game with magic-link authentication and a Postgres database secured with row-level security and schema migrations.
- **Simulation Engine:** Built a Deno Edge Function that runs the shared match-simulation engine through an RPC transaction to keep ranked results consistent.

HomeHelper — Graduate Multi-Agent Appliance-Repair Assistant 2025

Six-person graduate NLP team project; owned three of seven inference-time components

- **Orchestration and Training:** Built HomeHelper's orchestrator, Critic agent, and synthetic self-practice environment using repair corpora to iteratively improve a graduate multi-agent appliance-support system through automated feedback.
- **Multimodal Agent:** Built HomeHelper's video and image agent with a 4-bit vision-language model, Whisper transcription, three-level caching, and a React/WebSocket interface for multimodal appliance-repair assistance.

EduMUSE — Graduate Multi-Agent Tutoring Platform 2025

Four-person graduate ML team project; owned the architecture and the retrieval module

- **Architecture:** Owned EduMUSE's architectural foundation, building an EducationFlow abstract interface, a five-category FlowRegistry with structured fallback behavior, and Web, LLM, and Hybrid knowledge-retrieval workflows.
- **Evaluation:** Designed a five-metric weighted evaluation rubric covering relevance, credibility, coverage, recency, and accessibility; the hybrid workflow scored 9.15/10 versus 8.05 for LLM-only and 7.5 for web-only.

Teaching

Teaching Assistant 2024 – 2025

Baskin Engineering, University of California, Santa Cruz *Santa Cruz, CA*

- **Mentoring:** Mentored 150+ students across three courses and guided student project teams from design through delivery.
- **CSE 140, Artificial Intelligence:** Supported students through Pacman-based assignments on search, minimax and expectimax, reinforcement learning, and particle-filter inference, and helped maintain the course's assignment repository.
- **TIM 50 and TIM 58, Information Systems:** Guided project teams on databases, agile methods, and UML modeling in Business Information Systems and Systems Analysis and Design.
- **Course Operations:** Ran Gradescope autograders and assessment infrastructure across multiple terms.